SE LAB 3

COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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SECTION G

Problem Statement 51 : Virtual Technical Conference Platform

1. ARCHITECTURE SELECTION :

We chose Microservices Architecture for the Virtual Technical Conference Platform.
Microservices Architecture divides the system into small, independently deployable services. It supports independent scaling, fault isolation and independent deployment. However, it introduces operational complexity, network latency and data consistency challenges.

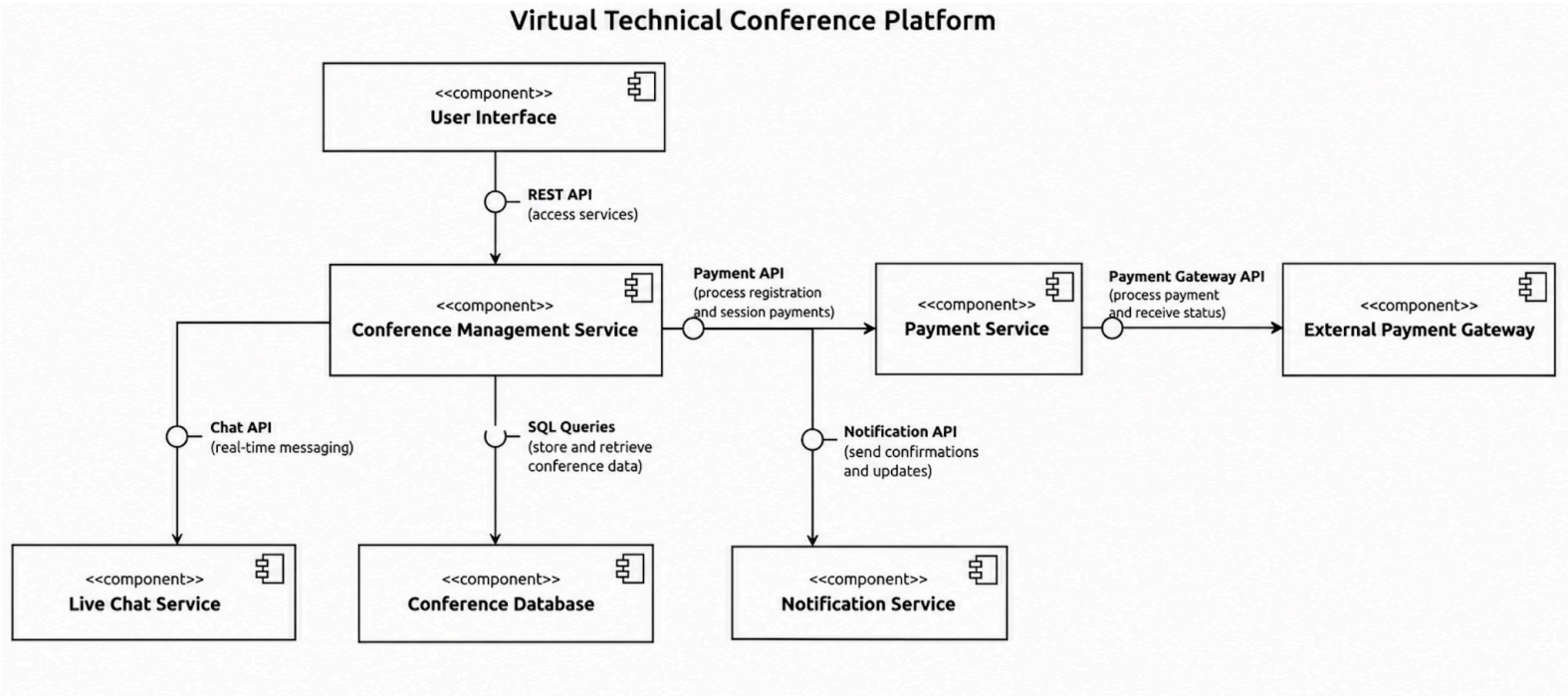
2. COMPONENT IDENTIFICATION :

The main components identified for the system are:

1. User Interface
2. Conference Management Service
3. Payment Service
4. External Payment Gateway
5. Live Chat Service
6. Conference Database
7. Notification Service

3. UML COMPONENT DIAGRAM :

Virtual Technical Conference Platform (Microservices Architecture)



4. INTERFACES AND COMPONENT INTERACTIONS :

REST API
Payment API
Payment Gateway API
Chat API
SQL Queries
Notification API

5. ARCHITECTURAL JUSTIFICATION :

Architectural Choice: Microservices Architecture

Reason 1 : Independent Scalability :
The platform must support live session chat for up to 500 concurrent users while delivering participant messages within 2 seconds. A separate Live Chat Service allows the chat functionality to be scaled independently according to the load without scaling the entire platform.

Reason 2 : Fault Isolation and Independent Deployment :
The platform contains different functionalities such as conference management, payment processing, live chat and notifications. Separating

these functionalities into services provides fault isolation and allows individual services to be updated or maintained independently.

Security Advantage :
Microservices allow security controls to be applied separately to different services. The Payment Service can securely communicate with the External Payment Gateway, while participant personal data and payment data can be protected using encryption in transit and at rest. This supports the security requirement of protecting sensitive participants and payment information.

Performance Benefit :
The Live Chat Service can be independently scaled to handle high numbers of concurrent users. This helps the system maintain the required message delivery time of less than 2 seconds under a load of 500 concurrent users.